The Regularized Rotational Vector Field (RRVF) Model: A Universal Amplitude Envelope and a Reproducible Numerical Audit

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Abstract

We study a three-component vector field on a one-dimensional periodic domain, governed by nonlinear advection, viscous diffusion, orthogonal rotation and nonlinear absorption damping. Applying the maximum principle to $q = \|T\|^2$ shows that the amplitude $M(t) = \max_x \|T\|$ is non-increasing and obeys the explicit envelope $M(t) \le M_0\,[1+2n\gamma M_0^{2n} t]^{-1/(2n)}$; letting $M_0 \to \infty$ gives a *universal* envelope $[2n\gamma t]^{-1/(2n)}$ that does not depend on the initial amplitude. The estimate controls the supremum norm only; it does not bound gradients, and it relies on the absence of a pressure term, so it does not address three-dimensional Navier–Stokes regularity. A deterministic finite-difference experiment (periodic centred differences, explicit Euler, a localised pulse at $t \approx 0.75$, $N_x = 200, 400, 800$) remains below the envelope. The discrete energy balance, including the pulse-induced jump, closes to round-off (2.5×10^{-15}) once the Euler increment $\frac{1}{2}\Delta t^2 \Delta x \sum |F|^2$ is included; without it the residual ($2.67 \times 10^{-2} \to 1.83 \times 10^{-3}$) is proportional to Δt. Successive-grid differences give exploratory apparent orders 1.91 (L^2) and 2.54 (vector L^∞) under coupled space–time refinement. All numbers are reproduced by the accompanying notebook.

Keywords: nonlinear damping; vector field; maximum principle; amplitude envelope; periodic finite differences; energy audit; grid refinement

1 Introduction and scope

A simple way to study amplitude control in nonlinear evolution equations is to isolate mechanisms that act directly on the field norm. The model considered here combines transport by one component of the field, viscous diffusion, a skew-symmetric (gyroscopic) rotation, and a nonlinear absorption term. The analysis concerns this specified reduced system only.

Relation to damped Navier–Stokes models. Absorption terms of the form $\alpha|u|^{\beta-1}u$ added to the three-dimensional incompressible Navier–Stokes equations have been studied as a stabilising mechanism: Cai and Jiu [4] proved global weak solutions for every $\beta \ge 1$ and global strong solutions for $\beta \ge 7/2$, and Zhong [5] showed that strong solutions exist globally for every $\beta \ge 1$. In that notation the damping used here corresponds to $\beta = 2n+1$. Those results concern a different system (with pressure and an incompressibility constraint) and are neither implied by, nor do they imply, the estimate below.

Contributions. (i) An explicit amplitude envelope, with a universal form independent of the initial amplitude and a proof that it is attained by spatially uniform solutions (Section 3). (ii) An exact discrete energy identity for explicit Euler, used to audit the simulation; with the Euler increment included, the

energy balance closes to round-off (Section 5). (iii) A fully specified, deterministic experiment with a notebook that reproduces every table and figure (Sections 4–7).

Scope. The unknown $T(x,t)$ has three real components and depends on one periodic coordinate; it is treated as a vector field. The term $\Omega \times T$ is an orthogonal gyroscopic coupling; no derivation from a higher-rank stress or torsion tensor is assumed. The name "regularized" refers to the presence of the absorption term. The system has no pressure and no divergence-free constraint, and it does not contain three-dimensional vortex stretching; it is therefore not the Navier–Stokes system, and no claim about the Navier–Stokes regularity problem [1, 2] is made.

2 Model

On the periodic interval $[-1,1)$ the field $T = (T_1, T_2, T_3)$ satisfies

$$\partial_t T + T_1\, \partial_x T = \nu\, \partial_{xx} T + \Omega \times T - \gamma \|T\|^{2n} T, \tag{1}$$

with $\nu \geq 0$, $\gamma > 0$, $n \geq 1$ and constant Ω. The simulations use $\Omega = (0,0,1)$, $\nu = 0.05$, $\gamma = 0.5$ and $n = 1$.

3 Analytical results

Proposition 1 (Norm identity)**.** *For a classical solution of* (1)*, $q = \|T\|^2$ satisfies*

$$\partial_t q + T_1\, \partial_x q = \nu\, \partial_{xx} q - 2\nu\, \|\partial_x T\|^2 - 2\gamma q^{n+1}. \tag{2}$$

Proof. Take the scalar product of (1) with $2T$. Then $2T \cdot \partial_t T = \partial_t q$, $2T_1\, T \cdot \partial_x T = T_1 \partial_x q$, $2T \cdot \partial_{xx} T = \partial_{xx} q - 2\|\partial_x T\|^2$, $T \cdot (\Omega \times T) = 0$, and $2T \cdot \gamma \|T\|^{2n} T = 2\gamma q^{n+1}$. □

Theorem 1 (Amplitude envelope)**.** *Let T be a classical solution on $[t_0, t_1] \times S^1$ (C^1 in t, C^2 in x) and let $M(t) = \max_x \|T(x,t)\|$ with $M(t_0) = M_0 > 0$. Then M is non-increasing and, for $t \in [t_0, t_1]$,*

$$M(t) \leq M_0 \Big[1 + 2n\gamma M_0^{2n}\, (t - t_0)\Big]^{-1/(2n)}. \tag{3}$$

Proof. The function $m(t) = M(t)^2 = \max_x q(x,t)$ is locally Lipschitz, because q is C^1 in t on a compact set. At every t where m is differentiable, Danskin's theorem gives $m'(t) = \partial_t q(x^*, t)$ for any maximiser x^*. At x^* we have $\partial_x q = 0$ and $\partial_{xx} q \leq 0$, so (2) yields

$$m'(t) \leq -2\nu \|\partial_x T\|^2 - 2\gamma m^{n+1} \leq -2\gamma m^{n+1} \leq 0.$$

Hence m is non-increasing; in particular, if M vanishes at some time it stays zero afterwards and (3) holds trivially. While $M > 0$, $M' \leq -\gamma M^{2n+1}$ almost everywhere, i.e. $\frac{d}{dt} M^{-2n} \geq 2n\gamma$. Integrating from t_0 gives $M(t)^{-2n} \geq M_0^{-2n} + 2n\gamma(t - t_0)$, which is (3). □

Corollary 1 (Universal envelope)**.** *Under the assumptions of Theorem 1, for every $t > t_0$ and* every *initial amplitude M_0,*

$$M(t) \leq \big[2n\gamma (t - t_0)\big]^{-1/(2n)}. \tag{4}$$

Remark 1 (Sharpness and role of the terms). The envelope depends only on γ, n and M_0; ν, the advection and Ω do not enter. It is attained by spatially uniform solutions: if T does not depend on x, then (2) reduces to $\dot{q} = -2\gamma q^{n+1}$ and (3) is an equality. Without damping ($\gamma = 0$) the same argument gives only $M(t) \leq M_0$: monotonicity of the amplitude is a structural property of the advection–diffusion–rotation part, and what the absorption term adds is the explicit, initial-amplitude-independent decay rate (4).

Remark 2 (Discontinuous-in-time perturbations). A pulse added at time t_{inj} is a discontinuity in time. Theorem 1 is applied on $[t_{\text{inj}}, t]$ with $t_0 = t_{\text{inj}}$ and M_0 the measured post-pulse amplitude.

Remark 3 (What is not controlled). The estimate bounds the supremum norm only. It gives no bound on $\partial_x T$ or on Sobolev norms. For $\nu = 0$ the system is hyperbolic and gradient steepening is not excluded by this argument.

Remark 4 (Why the argument does not extend to Navier–Stokes). For incompressible Navier–Stokes, the analogue of (2) contains the extra term $-2\,u \cdot \nabla p$, which is non-local and has no sign, so the maximum-principle step fails. A priori control of $\|u\|_{L^\infty}$ would imply regularity through classical criteria of Serrin type [6]; that control is precisely what the pressure term prevents us from obtaining in this way. Equation (1) has no pressure, which is why the argument closes.

4 Numerical method

Discretisation. The domain $[-1, 1)$ has period 2 and N_x uniformly spaced points, $\Delta x = 2/N_x$. Periodic centred differences approximate the derivatives, $(\partial_x T)_j \approx (T_{j+1} - T_{j-1})/(2\Delta x)$ and $(\partial_{xx} T)_j \approx (T_{j+1} - 2T_j + T_{j-1})/\Delta x^2$, and forward Euler advances the solution. Spatial integrals use rectangular quadrature.

Initial data and pulse.

$$T_1(x,0) = 3\exp\!\Big[-\tfrac{\sin^2(\pi x/2)}{0.15^2}\Big], \qquad T_2(x,0) = 1.5\exp\!\Big[-\tfrac{\sin^2(\pi(x-0.2)/2)}{0.12^2}\Big], \qquad T_3(x,0) = 0.$$

At the first time level with $t \geq 0.75$, the first component receives the periodic pulse

$$\Delta T_1(x) = 6\exp\!\Big[-\tfrac{\sin^2(\pi x/2)}{0.1^2}\Big].$$

Because injection occurs at the first time level meeting the threshold, its actual time t_{inj} depends on the grid (Table 1).

Time step. The step is $\Delta t = \min\big(0.25\,\Delta x^2/\nu,\ 0.4\,\Delta x/u_{\max}\big)$ with $u_{\max} = \max_x |T_1(x,0)| + 6 = 9$, then rescaled so that $t = 1$ is reached exactly. Three conditions are necessary for the linearised sub-problems of explicit Euler. (a) For $u_t + c u_x = \nu u_{xx}$ with centred differences, the amplification factor is $G(\theta) = 1 - 2d(1-\cos\theta) - i\lambda\sin\theta$ with $d = \nu\Delta t/\Delta x^2$ and $\lambda = c\Delta t/\Delta x$; one finds $|G| \leq 1$ for all θ if and only if $d \leq \tfrac{1}{2}$ and $\lambda^2 \leq 2d$, i.e. $\Delta t \leq \Delta x^2/(2\nu)$ and $\Delta t \leq 2\nu/c^2$. (b) The linearisation of $-\gamma\|T\|^{2n}T$ has eigenvalues $-\gamma\|T\|^{2n}$ and $-(2n+1)\gamma\|T\|^{2n}$, so $\Delta t \leq 2/\big((2n+1)\gamma M^{2n}\big)$. Table 1 reports the ratios $r_{\text{dif}} = \Delta t\,/\,(\Delta x^2/2\nu)$, $r_{\text{adv}} = \Delta t\,/\,(2\nu/u^2)$ with u the largest observed $|T_1|$, and $r_{\text{dmp}} = \Delta t\,/\,\big(2/(3\gamma M^2)\big)$ with M the largest observed post-pulse amplitude; all are below one. (c) Explicit Euler slightly amplifies the neutral rotation, by a factor $(1+\Omega^2\Delta t^2)^{1/2}$ per step, an $O(\Delta t)$ drift over a fixed horizon that is part of the Euler term of Section 5. These are conditions for frozen-coefficient linearisations [3]; they are not a stability proof for the fully discrete nonlinear scheme.

Table 1: Numerical setup. "Limiting" names the term that sets Δt; t_{inj} is the actual injection time; $r_{\text{dif}}, r_{\text{adv}}, r_{\text{dmp}}$ are the ratios defined in the text.

N_x	Δx	Δt	limiting	steps	t_{inj}	r_{dif}	r_{adv}	r_{dmp}
200	0.01	4.444×10^{-4}	advection	2250	0.750222	0.444	0.167	1.3×10^{-2}
400	0.005	1.250×10^{-4}	diffusion	8000	0.750000	0.500	0.048	3.6×10^{-3}
800	0.0025	3.125×10^{-5}	diffusion	32000	0.750000	0.500	0.012	9.1×10^{-4}

Reproducibility. The experiment is deterministic (no random numbers). The accompanying notebook `RRVF_audit.ipynb` regenerates all tables and figures in about fifteen seconds on a laptop (Python 3.12, NumPy 2.4, Matplotlib 3.10).

5 Discrete energy audit

The discrete energy is $E = \frac{\Delta x}{2}\sum_j \|T_j\|^2$. For $T^{k+1} = T^k + \Delta t\, F(T^k)$, with F the right-hand side of (1), the identity

$$E^{k+1} - E^k = \Delta t\, \Delta x \sum_j T_j^k \cdot F_j^k + \tfrac{1}{2}\Delta t^2\, \Delta x \sum_j \|F_j^k\|^2 \tag{5}$$

holds exactly. The audit accumulates the four physical rates $\Delta t\, \Delta x \sum T \cdot (\cdot)$ for diffusion, advection, rotation and damping, the pulse jump ΔE_{pulse} measured directly before and after injection, and the Euler term (second sum in (5)). The initial energy is $E(0) = 0.6480$ on every grid.

Table 2: Discrete energy audit: measured change and integrated rates (the rotation rate is zero up to round-off).

N_x	$E(1)$	ΔE measured	pulse jump	diffusion	advection	rotation	damping
200	0.2870	−0.361031	+1.6021	−1.0800	−0.0586	$\lesssim 5\times10^{-19}$	−0.8513
400	0.2878	−0.360262	+1.6023	−1.0661	−0.0619	$\lesssim 5\times10^{-19}$	−0.8419
800	0.2880	−0.360018	+1.6023	−1.0621	−0.0627	$\lesssim 5\times10^{-19}$	−0.8393

Table 3: Energy-balance closure. "Residual" is $|\Delta E - (\text{rates} + \text{jump})|$; the last column adds the Euler term.

N_x	Δt	rates + jump	residual	residual/Δt	Euler term	residual with Euler term
200	4.444×10^{-4}	−0.387772	2.674×10^{-2}	60.17	2.674×10^{-2}	2.5×10^{-15}
400	1.250×10^{-4}	−0.367617	7.355×10^{-3}	58.84	7.355×10^{-3}	1.8×10^{-15}
800	3.125×10^{-5}	−0.361844	1.826×10^{-3}	58.44	1.826×10^{-3}	2.3×10^{-15}

Two observations follow from Tables 2–3. First, the residual without the Euler term equals the Euler term to the digits shown, and the ratio residual/Δt stays between 58.4 and 60.2: the residual is a first-order-in-Δt discretisation defect, not exact conservation (Figure 1). Second, adding the Euler term closes the balance to 2.5×10^{-15} on all grids. Since (5) is an algebraic identity, this closure is a check that the bookkeeping is complete and correctly implemented; it is not an accuracy or convergence statement. The Euler term includes, among others, the contribution $\Omega^2(T_1^2 + T_2^2)$ created by the explicit treatment of the rotation. The rotation rate itself vanishes to 5×10^{-19} or less, as expected from $T\cdot(\Omega\times T) = 0$.

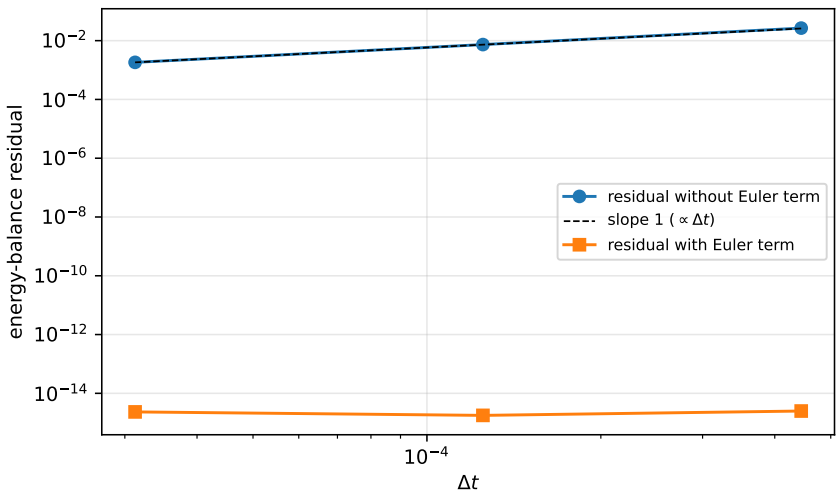


Figure 1: Energy-balance residual versus time step. Without the Euler term the residual follows the slope-one reference; with it, the residual is at round-off level.

6 Amplitude audit under a localised pulse

Table 4 compares the computed amplitude with the envelope (3), restarted from the measured post-pulse amplitude M_0 at the actual injection time, and with the universal envelope (4); the comparison is made at every step after injection. Figure 2 shows $M(t)$ and Figure 3 the profile of $\|T\|$ before the pulse, right after it, and at $t = 1$.

Table 4: Maximum-amplitude audit. "Min. margin" is the smallest value over the steps after injection of (restarted envelope $-M$); "respected" is the step-by-step comparison.

N_x	M_0 post-pulse	$M(1)$	restarted envelope	universal envelope	min. margin	respected
200	6.246590	1.200839	1.905520	2.000889	6.72×10^{-2}	yes
400	6.246800	1.201671	1.904758	2.000000	1.89×10^{-2}	yes
800	6.246840	1.202055	1.904759	2.000000	4.73×10^{-3}	yes

The computed amplitude stays below the envelope on all three grids ($M(1) \approx 1.201$ against 1.905). The minimum margin occurs at the first step after the pulse and equals about $151\,\Delta t$ on every grid: both curves coincide at t_{inj} by construction, and the computed amplitude then decays faster because the narrow pulse is also diffused. At $t = 1$ the margin is about 0.70. The universal envelope is close to the restarted one because M_0 is large: its value at $t = 1$ is 2.000. The envelope code is not specialised to $n = 1$: for $n = 2$ on $N_x = 400$ with $\Delta t = 2.5 \times 10^{-5}$, chosen so that the damping condition holds, one obtains $M_0 = 6.244$, $M(1) = 0.9139$ against an envelope of 1.1888, and an energy closure of 8.7×10^{-15}.

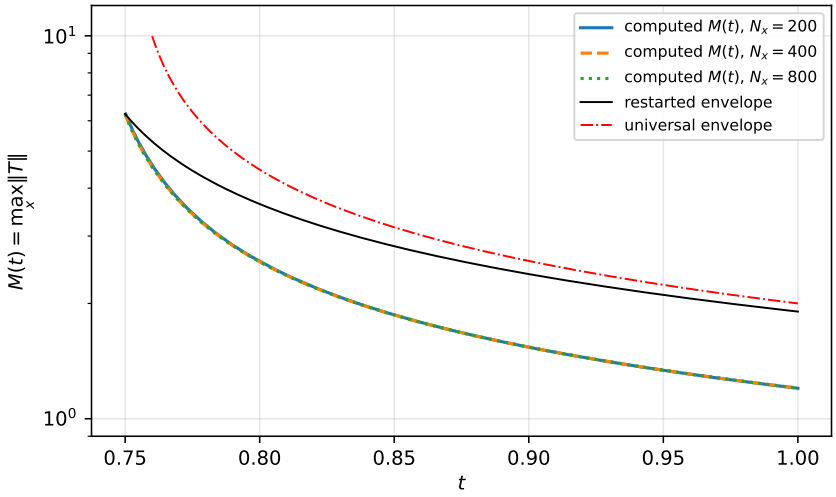


Figure 2: Computed amplitude $M(t) = \max_x \|T\|$ after the pulse on three grids, with the restarted envelope (3) and the universal envelope (4) ($N_x = 800$).

7 Grid refinement

Without an exact solution, successive-grid (Cauchy) differences are evaluated at coincident sample points at $t = 1$. The L^2 difference is $(\Delta x_c \sum_j \|\delta T_j\|^2)^{1/2}$ with Δx_c the coarser spacing, the vector L^∞ difference is $\max_j \|\delta T_j\|$, and the apparent order is $p = \log_2(D_{200,400}/D_{400,800})$.

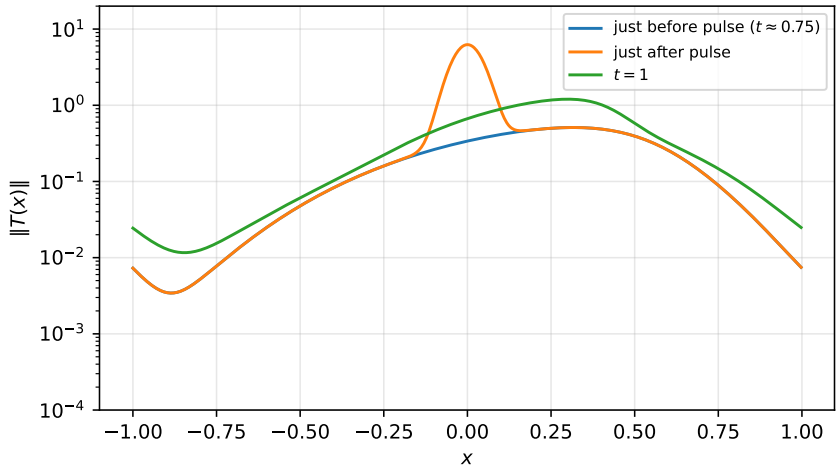


Figure 3: Profile of $\|T(x)\|$ for $N_x = 800$: just before the pulse, just after it, and at $t = 1$.

Table 5: Adjacent-grid differences at coincident points and apparent orders (exploratory).

norm	$D_{200,400}$	$D_{400,800}$	p
L^2	1.366×10^{-3}	3.637×10^{-4}	1.9092
L^∞ (vector)	3.219×10^{-3}	5.518×10^{-4}	2.5445

For explicit Euler with centred differences and $\Delta t \propto \Delta x^2$ an order close to two is expected, which is consistent with the L^2 value 1.91. The numbers must be read with care: space and time are refined together; $N_x = 200$ is advection-limited while $N_x = 400, 800$ are diffusion-limited; and the pulse is injected at $t_{\text{inj}} = 0.750222$ for $N_x = 200$ but at 0.75 on the two finer grids, which enters the first difference. With three levels, the value 2.54 in the supremum norm is best regarded as pre-asymptotic. The orders are exploratory, not established asymptotic rates, and $N_x = 800$ is a numerical reference, not an exact solution.

8 Limitations

- The envelope concerns classical solutions of the continuum system and bounds the supremum norm only; it gives no information on gradients or higher norms.
- The argument uses the absence of pressure and of an incompressibility constraint; it does not apply to Navier–Stokes.
- The time-step conditions are necessary for frozen-coefficient linearisations. Stability and convergence of the fully discrete nonlinear scheme are not proved; compliance with the envelope is observed, not proved, for the discrete solution.
- Space and time are refined together, so spatial and temporal errors are not separated. A study with fixed small Δt and varying Δx, and conversely, is the natural next step.
- One initial condition, one pulse and one parameter set are reported.

9 Conclusion

For the one-dimensional periodic vector model (1), the maximum principle gives a non-increasing amplitude and an explicit decay envelope; its universal form (4) bounds the amplitude after a time τ of

damping by $(2n\gamma\tau)^{-1/(2n)}$ whatever the amplitude was at the start, and it is attained by spatially uniform solutions. A finite-difference implementation with a localised pulse remains below the envelope on the three grids tested. The discrete energy balance, including the pulse, closes to round-off once the Euler increment is accounted for, which turns the earlier "discretisation defect" interpretation into an exact statement. Successive-grid differences decrease with exploratory apparent orders 1.91 and 2.54. Separating spatial and temporal errors, analysing the stability of the complete scheme, and establishing the physical interpretation of the model are left for future work.

Data and code availability

The notebook `RRVF_audit.ipynb` contains the complete implementation and the executed outputs. It is deterministic and reproduces every table and figure in this manuscript. A persistent identifier will be assigned upon deposit.

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